QUICK INSTALL · WINDOWS · ARDUINO IDE 2.X

CH32V003 in Arduino IDE

The hybrid install that actually gives you PWM and timers.

> **Why this is fiddly:** v1.0.4 (three years old) compiles and uploads, but has no `analogWrite()` and no `HardwareTimer`. That code exists only on the GitHub **master branch**, which was never released.
> So: install 1.0.4 to get the **toolchain**, then overwrite its **core files** with master.

Before you start

- Run Arduino IDE **as Administrator** for the first Boards Manager install — the WCH script creates symlinks and otherwise fails with `Cannot install tool WCH:beforeinstall`.
- Run `CH372DRV_S.exe` (from the WCH-LinkUtility ZIP) **before** plugging in the WCH-LinkE.
- **PD1 is the SWIO programming pin** — never repurpose it as GPIO.

The six steps

1. **Add the board URL** — File → Preferences → Additional Boards Manager URLs
   `https://github.com/openwch/board_manager_files/raw/main/package_ch32v_index.json`

2. **Install 1.0.4** — Boards Manager → search "CH32" → "CH32 MCU EVT Boards" 1.0.4
   This is what brings the RISC-V GCC compiler and upload tools. Skip it and you have no compiler.

3. **Close the IDE and back up the package folder** — copy to Desktop as 1.0.4-BACKUP
   `C:\Users\<YourName>\AppData\Local\Arduino15\packages\WCH\hardware\ch32v\1.0.4`

4. **Download master** — github.com/openwch/arduino_core_ch32 → Code → Download ZIP
   Extract it → folder `arduino_core_ch32-main`.

5. **Copy master over 1.0.4**
   Select everything inside the extracted folder, paste it into the 1.0.4 folder → **"Replace the files in the destination"**. That is `boards.txt`, `platform.txt`, `cores\`, `libraries\`, `variants\` — all of it. The folder still says 1.0.4; the IDE only reads what is inside.

6. **Reopen and pick the board**
   `Tools → Board → CH32V EVT Boards` → **CH32V003F4** (F4P6) or **CH32V00x**. Then `Tools → Clock` → **48 MHz** — that Clock entry only appears after the overwrite, so it doubles as proof.

Verify

```
// PD2 = LED on your board (also a default PWM pin)
#define LED PD2

void setup() { pinMode(LED, OUTPUT); }

void loop() {
  for (int b = 0; b <= 255; b += 5) { analogWrite(LED, b); delay(20); }
  for (int b = 255; b >= 0; b -= 5) { analogWrite(LED, b); delay(20); }
}
```

Smooth fade = master-branch PWM is live. Hard on/off blink = still on the stale core. Keep the clock at 48 MHz (that is what NeoPixel/WS2812 bit-bang timing assumes).

When it goes wrong

Board missing from menu	`boards.txt` must sit *directly* in the 1.0.4 folder — no extra nested folder from unzipping.
An "update" broke it	Expected: Board Manager sees 1.0.4 and overwrites master. Reinstall, redo step 5. (Or drop the core in `Documents\Arduino\hardware\WCH\ch32v\` — no version subfolder — and it lists as "in sketchbook", invisible to updates.)
Upload fails	Reinstall `CH372DRV_S.exe`; check `Tools → Upload Method` is WCH-LinkE.
PC1 won't PWM	Not a default PWM pin — TIM2_CH4 only after an alternate-function remap. Use **PD2** or **PA0**.
WS2812 garbage	NeoPixel only works at 48 MHz. Power the strip from 5 V, with 330–470 Ω in series on data.

Order matters: Boards Manager 1.0.4 → close IDE → overwrite with master → restart → select CH32V003 + 48 MHz → code.

Condensed from the full CH32V003 setup guide · verified against openwch/arduino_core_ch32 and community docs.